

1 Evan-krylov theorem2(CC)

Thm6.1

Let F be concave and let $u \in C^2(B_1)$ satisfy

$$F(D^2u) = 0 \quad \text{in } B_1.$$

Then $u \in C^{2,\alpha}(B_{1/2})$ and

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \leq C\|u\|_{C^{1,1}(B_1)}.$$

Lemma6.2 Under the hypothesis of Theorem 6.1, there exists a universal constant $0 < \delta < 1$ such that

$$\text{osc}_{B_\delta} D^2u \leq \frac{1}{2} \text{osc}_{B_1} D^2u = 1.$$

Oscillation decay method

$\delta^k x \in B_{\delta^k}$ if $x \in B_1, w_k(x) = u(\delta^k x)$ and $\delta^{-2k} D^2 w_k(x) = D^2 u(\delta^k x)$.

$$\delta^{2k} F(\delta^{-2k} D^2 w_k(x)) = 0 \quad \text{in } B_1.$$

Then

$$\text{osc}_{B_{\delta^2}} D^2u(x) = \text{osc}_{B_\delta} D^2u(\delta x) \leq \frac{1}{2} \text{osc}_{B_1} D^2u(\delta x) = \frac{1}{2} \text{osc}_{B_\delta} D^2u(x).$$

$$\text{osc}_{B_{\delta^3}} D^2u(x) = \text{osc}_{B_\delta} D^2u(\delta^2 x) \leq \frac{1}{2} \text{osc}_{B_1} D^2u(\delta^2 x) = \frac{1}{2} \text{osc}_{B_{\delta^2}} D^2u(x).$$

Can get

$$\text{osc}_{B_{\delta^k}} D^2u(x) \leq \frac{1}{2} \text{osc}_{B_{\delta^{k-1}}} D^2u(x) \leq \frac{1}{2^{k-1}} \text{osc}_{B_\delta} D^2u(x) \leq 2^{1-k}.$$

Let $x \in B_{1/2}$ and let k be such that $x \in B_{\delta^k}/B_{\delta^{k+1}}$. Then,

$$|D^2u(x) - D^2u(0)| \leq \text{osc}_{B_{\delta^k}} D^2u(x) \leq 2^{1-k}.$$

According to the selection of k with $\delta^k \geq |x| \geq \delta^{k+1}$, $\frac{\log|x|}{\log\delta} \geq k \geq \frac{\log|x|}{\log\delta} - 1$, then

$$|D^2u(x) - D^2u(0)| \leq 2^{1-k} \leq 2 \cdot 2^{-\frac{\log|x|}{\log\delta} + 1} = 4 \cdot |x|^{-\frac{\log 2}{\log\delta}}.$$

Finally $\alpha = -\frac{\log 2}{\log\delta} > 0$,

$$|D^2u(x) - D^2u(0)| \leq 2^{1-k} \leq C|x|^\alpha.$$

Lemma 6.3

Let $v \in \overline{S}(\lambda, \Lambda, 0)$ in B_1 satisfy $v \geq 0$ in B_1 . Then

$$\inf_{B_{1/2}} v \geq C |\{v \geq 1\} \cap B_{1/4}|^\delta,$$

where C and δ are positive universal constants.

Weak Harnack inequality $p = p(n, \lambda, \Lambda)$.

$$\begin{aligned} \inf_{B_{1/2}} v &\geq C \|v\|_{L^p(B_{1/4})} = C \left[\int_{B_{1/4}} v^p dx \right]^{\frac{1}{p}} = C \left[\int_{\{v \geq 1\} \cap B_{1/4}} v^p dx \right]^{\frac{1}{p}} \\ &\geq C \left[\int_{\{v \geq 1\} \cap B_{1/4}} 1^p dx \right]^{\frac{1}{p}} = C |\{v \geq 1\} \cap B_{1/4}|^{\frac{1}{p}}. \end{aligned}$$

Lemma 6.4 $F(M_1) = F(M_2) = 0$, then

$$c_0 \|(M_1 - M_2)\| \leq \|(M_1 - M_2)^+\| = \sup_{e \in \mathbb{R}^n, |e|=1} |(M_1 - M_2)^+ e \cdot e|,$$

where $c_0 = \lambda/(\lambda + \Lambda)$.

$$\begin{aligned} 0 = F(M_2) &\leq F(M_1) + \Lambda \|(M_1 - M_2)^+\| - \lambda \|(M_1 - M_2)^-\| \\ &\leq (\Lambda + \lambda) \|(M_1 - M_2)^+\| - \lambda \|(M_1 - M_2)^-\| \end{aligned}$$

For $N \geq 0$, and $N = N^+ - N^-$ where $N^+ \geq 0$, $N^- \geq 0$.

$$\lambda \|N\| \leq F(M + N) - F(M) \leq \Lambda \|N\|.$$

分别调整 M 和 N ,

$$F(M + N + N^-) - F(M + N) \geq \lambda \|N^-\|,$$

$$F(M + N^+) - F(M + N) \geq \lambda \|N^-\|,$$

$$F(M + N^+) - F(M) \leq \Lambda \|N^+\|,$$

finally,

$$F(M + N) - F(M) \leq \Lambda \|N^+\| - \lambda \|N^-\|.$$

Lemma 6.5 Under the hypothesis of Theorem 6.1, assume that

$$1 < \text{diam } D^2 u(B_1) \leq 2$$

and that $D^2 u(B_1)$ is covered by N balls $B^1, \dots, B^N$ of radius ε (in the space $\mathcal{S}$ of symmetric matrices) with $N \geq 1$ and $\varepsilon \leq \varepsilon_0$, where $\varepsilon_0 > 0$ is a universal constant. Then $D^2 u(B_{1/2})$ is covered by $N - 1$ balls among $B^1, \dots, B^N$.

Proof of Lemma 6.5 If $\exists i \in \{1, 2, 3, \dots, N\}$ such that $B^i \cap D^2u(B_1) = \emptyset$ and since $D^2u(B_1) \subset \bigcup_{k=1}^N B^k$, can get

$$D^2u(B_{1/2}) \subset D^2u(B_1) \subset \bigcup_{k \neq i}^N B^k.$$

Otherwise, $\forall i \in \{1, 2, 3, \dots, N\}$ such that $B^i \cap D^2u(B_1) \neq \emptyset$. Then for every B^i , $\exists x_i \in B_1$ such that $M_i = D^2u(x_i) \in B^i$. M_i is a point in B^i . So

$$B^i \subset B_{2\varepsilon}(M_i).$$

$C_o = \frac{\lambda + \Lambda}{\Lambda}$ and taking ε_o such that $2\varepsilon \leq 2\varepsilon_o \leq C_o/16$. So

$$D^2u(B_1) \subset \bigcup_{k=1}^N B^k \subset \bigcup_{i=1}^N B_{2\varepsilon}(M_i) \subset \bigcup_{i=1}^N B_{C_o/16}(M_i).$$

Exist one ball $\overline{B}$ of radius 2 in $\mathcal{S}$.

$$\bigcup_{i=1}^N M_i \subset \overline{B}$$

If $\|M_1 - M_2\| \leq C_o/16$ then $B_{C_o/16}(M_1) \subset B_{C_o/8}(M_2)$, can replace

$$D^2u(B_1) \subset \bigcup_{i=1}^N B_{C_o/16}(M_i) \subset B_{C_o/8}(M_2) \cup \left\{ \bigcup_{i=3}^N B_{C_o/16}(M_i) \right\}.$$

Set

$$N' = \{n | \text{dist}(M_i, M_j) \geq C_o/16, 1 \leq i, j \leq n, i \neq j\}.$$

Then

$$D^2u(B_1) \subset \bigcup_{i=1}^{N'} B_{C_o/8}(M_i),$$

$$B_1 \subset \bigcup_{i=1}^{N'} (D^2u)^{-1} B_{C_o/8}(M_i).$$

Therefore, there exists one M_i say M_1 , such that

$$|(D^2u)^{-1}(B_{C_o/8}(M_1)) \cap B_{1/4}| \geq \eta > 0.$$

And take $2\varepsilon \leq 2\varepsilon_o \leq 1/4$ again.

$$D^2u(B_1) \subset \bigcup_{i=1}^N B_{2\varepsilon}(M_i),$$

and $\text{diam} D^2u(B_1) > 1$, $\exists M_2$ such that $\|M_1 - M_2\| > 1/4$. Otherwise $\|M_1 - M_i\| < 1/4, i = 2, 3, \dots, N$. Can $\forall y \in \bigcup_{i=1}^N B_{2\varepsilon}(M_i)$, then $y \in B_{2\varepsilon}(M_i)$, then

$\|y - M_1\| \leq \|y - M_i\| + \|M_i - M_1\| \leq 2\varepsilon + 1/4 \leq 1/2$, then $\text{diam} \bigcup_{i=1}^N B_{2\varepsilon}(M_i) \leq 1$. But

$$D^2u(B_1) \subset \bigcup_{i=1}^N B_{2\varepsilon}(M_i),$$

contradiction with $\text{diam} D^2u(B_1) > 1$. From Lemma 6.4

$$c_0/4 < c_0\|(M_1 - M_2)\| = \sup_{e \in \mathbb{R}^n, |e|=1} (M_1 - M_2)^+ e \cdot e = \sup_{e \in \mathbb{R}^n, |e|=1} (u_{ee}(x_1) - u_{ee}(x_2))^+,$$

then $\exists e \in \mathbb{R}^n, |e| = 1$ such that $u_{ee}(x_1) - u_{ee}(x_2) > c_0/4$.

u satisfy $F(D^2u) = 0$, then $u_{ee} \in \underline{S}(\frac{\lambda}{n}, \Lambda, 0)$, set $K = \sup_{B_1} u_{ee}$, $v = K - u_{ee}$, then $v \in \overline{S}(\frac{\lambda}{n}, \Lambda, 0)$ and $v > c_0/4$. $\forall x \in (D^2u)^{-1}(B_{c_0/8}(M_1))$ can get $u_{ee}(x) < u_{ee}(x_1) + c_0/8$ and $K \geq u_{ee}(x_1) + c_0/4$, finally

$$v = K - u_{ee}(x) \geq u_{ee}(x_1) + c_0/4 - (u_{ee}(x_1) + c_0/8) = c_0/8.$$

So $(D^2u)^{-1}(B_{c_0/8}(M_1)) \subset \{v \geq c_0/8\}$.

$$|\{v \geq c_0/8\} \cap B_{1/4}| \geq \eta > 0.$$

$$|\{8v/c_0 \geq 1\} \cap B_{1/4}| \geq \eta > 0.$$

From Lemma 6.3 have

$$\inf_{B_{1/2}} (K - u_{ee}) \geq C_1 > 0.$$

From

$$D^2u(B_1) \subset \bigcup_{i=1}^N B_{2\varepsilon}(M_i),$$

get

$$u_{ee}(B_1) \subset \bigcup_{i=1}^N e^t B_{2\varepsilon}(M_i) \subset \bigcup_{i=1}^N B_{2\varepsilon}(u_{ee}(x_i)),$$

take sup, $\exists \hat{i}$ such that

$$K \in B_{2\varepsilon}(u_{ee}(x_{\hat{i}})), K - u_{ee}(x_{\hat{i}}) < 3\varepsilon.$$

Finally, take $5\varepsilon \leq 5\varepsilon_0 \leq C_1$,

$$K - u_{ee} \geq C_1 \geq 5\varepsilon \quad \text{in} \quad B_{1/2},$$

so $\text{dist}(u_{ee}(x_{\hat{i}}), u_{ee}(B_{1/2})) > 2\varepsilon$

$$D^2u(B_{1/2}) \cap B_{2\varepsilon}(M_{\hat{i}}) = \emptyset$$

$$D^2u(B_{1/2}) \cap B^{\hat{i}} = \emptyset$$

proof of Lemma 6.2 $\text{osc}_{B_1} D^2 u = 2$, then $D^2 u(B_1)$ is bounded. Then exist N balls covers $D^2 u(B_1)$, from Lemma 6.5 $D^2 u(B_{1/2})$ is covered $N - 1$ balls. Set $w(x) = 4u(x/2)$, and $x \in B_1$, then

$$\text{osc}_{B_1} D^2 w = \text{osc}_{B_{1/2}} D^2 u = \text{osc}_{B_1} D^2 u \leq 2,$$

$1 < D^2 w(B_1) \leq 2$ is covered $N - 1$ balls, then $D^2 w(B_{1/2})$ is covered $N - 2$ balls.

If

$$\text{osc}_{B_{1/2^{N-1}}} D^2 u > 1,$$

then $D^2 u(B_{1/2^{N-1}})$ is covered 1 balls, $D^2 u(B_{1/2^N})$ is covered 0 balls, that means $D^2 u(B_{1/2^N}) = \emptyset$, contradiction.